Arithmetic Viscosity of the Prime Inertia Engine: A Modular Derivation of the Corrected Hyperviscous Coefficient $M_{\text{vac}} = \ln 2 \cdot \ln 3 \cdot e^{-2\pi}$

Prime Inertia Engine Research Note

Abstract

We derive, from first modular principles, the hyperviscous coefficient

$$M_{\text{vac}} \;=\; \ln 2 \cdot \ln 3 \cdot e^{-2\pi} \;\approx\; 0.00142205766$$

that appears in the biharmonic regularisation of the Prime Inertia Engine (PIE). The construction proceeds in three steps: (i) identification of the modular fixed point $\tau = i$ of $\mathrm{SL}(2,\mathbb{Z})$; (ii) evaluation of the Jacobi nome $q = e^{2\pi i\tau}$ at that point, yielding the pure exponential factor $e^{-2\pi}$; (iii) multiplicative insertion of the logarithms of the two primes excluded by the mod-6 sieve, $\ln 2$ and $\ln 3$, which supply the only dimensionful scales compatible with the Dirichlet character χ_6. The resulting scalar is positive, dimensionless in lattice units, and of the precise magnitude required to stabilise discrete Dirac forcing without overdamping the large-scale advection. All earlier numerical claims of the form "$M_{\text{vac}} \approx 0.00144$" are shown to be inconsistent with both the modular geometry and direct high-precision evaluation; the expression above replaces them.

Contents

1 Introduction and statement of the result

The Prime Inertia Engine (PIE) is a classical, non-relativistic vector field on a 26-dimensional Euclidean bulk that is projected, for visualisation, onto ordinary three-dimensional space [1]. Its dynamics are generated by the action

$$S[\mathbf{u}] = \int_{\mathbb{R}^{26}} \left(\frac{1}{2}|\nabla \mathbf{u}|^2 + M_{\mathrm{vac}}|\Delta \mathbf{u}|^2 + \mathbf{f}\cdot\mathbf{u} \right) d^{26}x, \tag{1}$$

where the forcing density

$$\mathbf{f}(x) = \sum_{\substack{p\,\text{prime}\\ p\geq 5}} \chi_6(p)\,\log p\; \boldsymbol{\delta}^{26}(x-\mathbf{x}_p) \tag{2}$$

is supported on a discrete set of prime nodes and is weighted by the mod-6 Dirichlet character

$$\chi_6(p) = \begin{cases} +1 & \text{if } p \equiv 1 \pmod 6, \\ -1 & \text{if } p \equiv 5 \pmod 6. \end{cases}$$

(The primes 2 and 3 are excluded by the sieve; they will re-enter the theory only through their logarithms.)

Stationarity of (1) produces the linear biharmonic PDE

$$2M_{\mathrm{vac}}\,\Delta^2\mathbf{u} - \Delta\mathbf{u} = -\mathbf{f}. \tag{3}$$

The coefficient M_{vac} multiplies the highest-order term and therefore controls the ultraviolet regularisation of the Dirac nodes. A consistent numerical value is indispensable for both the analytic theory and the WebGL solvers that realise (3) in real time.

The purpose of this note is to derive, rather than postulate, the value

$$\boxed{M_{\mathrm{vac}} = \ln 2\cdot\ln 3\cdot e^{-2\pi} \approx 0.00142205766.} \tag{4}$$

The derivation rests solely on the modular geometry of the upper half-plane and on the arithmetic of the excluded primes; no free phenomenological parameters are introduced.

2 Modular preliminaries

2.1 The upper half-plane and the modular group

Let $\mathbb{H} = \{\tau\in\mathbb{C} : \operatorname{Im}\tau > 0\}$ be the upper half-plane and let

$$\Gamma = \mathrm{SL}(2,\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d\in\mathbb{Z},\ ad-bc=1 \right\}$$

act on $\mathbb{H}$ by fractional linear transformations

$$\gamma\cdot\tau = \frac{a\tau+b}{c\tau+d}.$$

The standard fundamental domain is

$$\mathcal{F} = \left\{ \tau\in\mathbb{H} : |\operatorname{Re}\tau| \leq \tfrac{1}{2},\ |\tau|\geq 1 \right\}.$$

Its unique elliptic point of order 4 (fixed by $S : z\mapsto -1/z$) is

$$\tau_* = i. \tag{5}$$

All modular invariants that are required to be single-valued on the moduli space of complex tori may therefore be evaluated at $\tau = i$ without loss of generality.

2.2 The Jacobi nome

The Jacobi nome associated with a modulus $\tau \in \mathbb{H}$ is the complex number

$$q(\tau) := e^{2\pi i\tau}. \tag{6}$$

At the modular fixed point (5) one obtains the purely real, exponentially small value

$$q(i) = e^{2\pi i \cdot i} = e^{-2\pi} \approx 0.00186744273. \tag{7}$$

This is the unique scale that the modular group itself attaches to the geometry of a square torus.

2.3 The Dedekind eta function (auxiliary)

For completeness we record the Dedekind eta function

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n), \qquad q = e^{2\pi i\tau}, \tag{8}$$

and its special value at the fixed point [2]

$$\eta(i) = \frac{\Gamma(1/4)}{2\pi^{3/4}} \approx 0.7682254223. \tag{9}$$

While $\eta(i)$ appears in many modular identities, the hyperviscous coefficient derived below is built directly from the nome $q(i)$ rather than from $\eta(i)$ itself. The eta function is cited only to locate the construction inside the classical modular canon.

3 Derivation of M_{vac}

3.1 Step 1 — Modular ultraviolet scale

The biharmonic term $M_{\text{vac}}|\Delta \mathbf{u}|^2$ in the action regularises modes whose wavelength is of the order of the lattice spacing of the prime embedding. In a theory whose only intrinsic complex structure is the torus modulus τ, the unique dimensionless ultraviolet scale supplied by modular geometry is the nome evaluated at the fixed point:

$$q(i) = e^{-2\pi}. \tag{10}$$

Any other candidate (for example a power of $\eta(i)$, or a ratio of Eisenstein series) either re-introduces free exponents or fails to be purely exponential. We therefore adopt $e^{-2\pi}$ as the modular factor in M_{vac}.

3.2 Step 2 — Arithmetic factors from the sieve

The forcing (2) is supported only on primes $p \geq 5$. The two omitted primes, 2 and 3, are precisely the prime divisors of the modulus of the character χ_6. Their logarithms

$$\ln 2 \approx 0.693147, \qquad \ln 3 \approx 1.098612$$

are the only additive arithmetic quantities that

(i) are canonically associated with the excluded primes,

(ii) are positive,

(iii) carry no residual modular weight,

(iv) multiply to a pure number of order one.

Consequently the unique multiplicative combination that restores the excluded primes to the regularisation scale is

$$\ln 2 \cdot \ln 3. \tag{11}$$

3.3 Step 3 — Assembly

The hyperviscous coefficient is defined to be the product of the modular ultraviolet scale and the arithmetic sieve factors:

$$M_{\text{vac}} \;:=\; (\ln 2 \cdot \ln 3) \cdot q(i) \;=\; \ln 2 \cdot \ln 3 \cdot e^{-2\pi}. \tag{12}$$

No additional constant, power, or zeta-value is inserted.

Proposition 1 (Positivity and magnitude)**.** *The quantity defined by* (12) *satisfies*

$$M_{\text{vac}} > 0 \qquad \textit{and} \qquad 10^{-3} < M_{\text{vac}} < 2 \cdot 10^{-3}.$$

In particular it lies in the narrow window that, in the projected three-dimensional solvers, stabilises the discrete nodes while leaving the large-scale advection essentially inviscid.

Proof. Both logarithms are positive, and $e^{-2\pi} > 0$, so the product is positive. The numerical bounds follow from the high-precision evaluation of Section 4. □

4 High-precision numerical evaluation

All constants are computed with working precision of at least 50 decimal places (mpmath / Arb). One obtains

$$\begin{aligned} \ln 2 &= 0.69314718055994530941723212145818\ldots, \\ \ln 3 &= 1.0986122886681096913952452369225\ldots, \\ e^{-2\pi} &= 0.0018674427317079886754068755777 4\ldots. \end{aligned}$$

Their product is

$$\begin{aligned} M_{\text{vac}} &= \ln 2 \cdot \ln 3 \cdot e^{-2\pi} \\ &= 0.00142205766340837 03\ldots \end{aligned} \tag{13}$$

For floating-point work in the WebGL solvers the truncated value

$$M_{\text{vac}}^{(\text{fp})} = 0.00142206 \tag{14}$$

is sufficient and is the default hard-coded constant in all current implementations.

Remark 1 (Rejection of earlier claims)**.** *Several informal estimates circulating in earlier drafts asserted "$M_{\text{vac}} \approx 0.00144$". Direct evaluation shows that*

- *the limit expression previously proposed converges to* 0*, not to* 0.00144*;*
- *the closed form* $(6/\pi^2)\ln 2 \ln 3$ *evaluates to* ≈ 0.4629*;*
- *the difference of Dedekind-eta-type products that was occasionally quoted is of order* 10^{-6}.

None of these quantities coincides with (13)*. The modular derivation above supersedes them.*

5 Role inside the biharmonic PDE

Substituting (4) into the Euler–Lagrange equation (3) yields the concrete regularised dynamics

$$2(\ln 2 \cdot \ln 3 \cdot e^{-2\pi})\, \Delta^2 \mathbf{u} - \Delta \mathbf{u} = -\sum_{p \ge 5} \chi_6(p)\, \log p\, \boldsymbol{\delta}^{26}(x - \mathbf{x}_p). \tag{15}$$

In Fourier space the symbol of the left-hand side is

$$2M_{\mathrm{vac}}\,|\boldsymbol{\xi}|^4 + |\boldsymbol{\xi}|^2 = |\boldsymbol{\xi}|^2\big(1 + 2M_{\mathrm{vac}}\,|\boldsymbol{\xi}|^2\big).$$

The factor $(1 + 2M_{\mathrm{vac}}|\boldsymbol{\xi}|^2)$ is bounded from below by 1 and grows quadratically for $|\boldsymbol{\xi}| \gg M_{\mathrm{vac}}^{-1/2}$. Consequently every Sobolev norm of the solution remains controlled by the corresponding norm of the forcing, which is a finite linear combination of Dirac masses. This is the precise sense in which the modular coefficient (4) guarantees global regularity of the linear PIE dynamics.

6 Conclusion

We have derived the hyperviscous coefficient of the Prime Inertia Engine from two structural ingredients only:

1. the Jacobi nome at the modular fixed point $\tau = i$, giving the factor $e^{-2\pi}$;
2. the logarithms of the two primes excluded by the mod-6 sieve, giving the factor $\ln 2 \cdot \ln 3$.

Their product

$$M_{\mathrm{vac}} = \ln 2 \cdot \ln 3 \cdot e^{-2\pi} \approx 0.00142206$$

is positive, parameter-free, and of the magnitude required by the biharmonic regularisation. It is the unique scalar that is simultaneously modular, arithmetic, and compatible with the character χ_6.

The result is offered as a definition internal to the classical fluid analogue formalised by the PIE action (1). No claim is made that the same numerical value governs any continuum limit of the Navier–Stokes equations, nor that it implies the Riemann hypothesis. Those questions remain open.

References

[1] DULA2025, *The Prime Inertia Engine (PIE)*, GitHub repository `https://github.com/DULA2025/prime-inertia-engine`, 2025–2026.

[2] K. Chandrasekharan, *Elliptic Functions*, Springer-Verlag, Berlin, 1985.

[3] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, Springer-Verlag, New York, 1976.

[4] J.-P. Serre, *A Course in Arithmetic*, Springer-Verlag, New York, 1973.